

Problem

Given a string containing lowercase English letters, find and print all the characters that occur more than once.

You must use bit manipulation to track the characters that have already appeared.

Input Format

The first line contains a string S

Constraints

$1 \leq |S| \leq 10^5$ S contains only lowercase English letters (a-z)

Output Format

Print all duplicate characters in the order of their first repeated occurrence. If there are no duplicates, print No duplicates.

Sample Input 0

programming

Sample Output 0

r g m

Sample Input 1

greater

Sample Output 1

r e

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Language

Java 7



```
1 import java.io.BufferedReader;
2 import java.io.InputStreamReader;
3 import java.io.IOException;
4 import java.util.ArrayList;
5 import java.util.List;
6
7 public class Solution {
8
9     public static void printDuplicates(String s)
10         int seenOnce = 0;
11         int seenMultiple = 0;
12
13         for (int i = 0; i < s.length(); i++) {
14             char ch = s.charAt(i);
15             int bitPosition = ch - 'a';
16
17             if ((seenOnce & (1 << bitPosition))
18                 seenMultiple |= (1 << bitPosition);
19         }
20     }
```

Line: 63 Col: 2



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success